INCH-POUND
MIL-DTL-24643/15E
22 August 2002
SUPERSEDING
MIL-C-24643/15D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE LSDSGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualifications required.

Construction (watertight with circuit integrity)
Sizes 3 to 100 inclusive, extruded silicone rubber insulation 1/

First - Copper conductor, uncoated, (see table I for size).
Second - Extruded silicone rubber insulation (see table I for diameter).
Third - Glass braid.
Fourth - Braid covering (see table I for diameter). Standard identification code applied by method 1 or method 3.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:58Z
Check the source to verify that this is the current version before use.

Sizes 23 to 400 inclusive, silicone rubber-glass tape insulation [1/]

First - Copper conductor, uncoated, (see table I for size)
Second - Silicone rubber - glass tape insulation (see table I for diameter). Letter identification code applied by method 5 or standard identification code method 1.
Third - The two conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly.
Fourth - A binder.

1/ Size 23 through 100 may be glass tape or extruded silicone rubber insulation.

Sizes 3 to 100 inclusive, extruded silicone rubber insulation [1/]

Fifth - The two conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly
Sixth - Binder or combination binder and barrier.
Seventh - Cross-linked polyolefin jacket. (see table I for thickness).

Sizes 23 to 400 inclusive, silicone rubber-glass tape insulation [1/]

Fifth - Cross-linked polyolefin jacket. (see table I for thickness).

1/ Size 23 through 100 may be glass tape or extruded silicone rubber insulation at the manufacturer's option.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:58Z
Check the source to verify that this is the current version before use.

TABLE I. Details.

Military part no. M24643/15	Size	Conductor size		Type's overall diameter (inches)		Diameter over outer insulation outer tape nom. (inch)	Diameter over braid maximum (inch)	Diameter over Insulation minimum (inch)	Jacket thickness minimum average (inch)	Conductor resistance (ohms)	Insulation resistance (megohms)	Cold bending mandrel diameter maximum	Accelerated service loading ampere
		Navy std	AWG	LSDGU									
				min	max								
-01UN	3		16 (Cl B)		0.391		0.130	0.096	0.030	4.3	500		
-02UN	4		14 (Cl B)		0.427		0.143	0.112	0.030	2.68	500		
-03UN	9		10 (Cl B)		0.544	----	0.187	0.154	0.040	1.08	500	----	----
-04UN	14		9 (Cl B)		0.670		0.262	0.230	0.040	0.859	500		100
-05UN	23		7 (Cl B)		0.781	0.274	0.310	0.271	0.050	0.543	500		130
-06UN	50		3 (Cl C)		0.911	0.334	0.380	0.335	0.050	0.21	200	12	215
-07UN	75		1 (Cl C)		1.074	0.407	0.440	0.407	0.050	0.134	175	13	285
-08UN	100		0 (Cl D)		1.167	0.453	0.500	0.453	0.050	0.106	160	15	335
-09UN	200		0000 (Cl D)		1.583	0.634			0.060	0.053	125	20	525
-10UN	300		300 MCM (Cl D)		1.841	0.748			0.075	0.0377	110	23	700
-11UN	400	400 (127)			2.069	0.862			0.075	0.0273	100	25	835

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:58Z
Check the source to verify that this is the current version before use.

EXAMINATION AND TESTS

Basic electrical:	Requirements:
Conductor resistance - ohms/1000 feet at 25°C, maximum	(see table I)
Voltage withstand - volts, root mean square, minimum	
Sizes 3 through 9	3000
Sizes 14 and larger	5000
Insulation resistance - conductor to conductor,	
Megohms -1000 feet, minimum	(see table I)
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-DTL-24643 for limits of water leakage	No failure
Abrasion resistance - sizes 3 through 100 (extruded	
silicone rubber) scrapes, minimum	250
Crack resistance - sizes 3 through 100 (extruded silicone rubber)	No damage
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Gas flame - 1 hour	No failure
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Cold bending, cable - (see table I for mandrel diameter)	No damage
Physicals (unaged)	
Insulation (extruded)	
Tensile strength - lb/in^2, minimum	750
Elongation - percent, minimum	125
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
Permanence of printing (conductor, method 1 only) cycles, minimum	
(extruded silicon rubber only)	25
Permanence of printing (jacket) - cycles, minimum	125

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:58Z
Check the source to verify that this is the current version before use.

Cable sealant removability..... No failure

Group D:

Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements:
Cold working (minus 20 ± 2°C)	No damage
Gas flame (3 hours)	No failure
Specific gravity of extruded silicone insulation, maximum	1.55
Accelerated service - sizes 14 and larger	
(see table I for load current)	No failure
Aging and compatibility (cable) (125 ± 5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSDSGU-3	2500
LSDSGU-4 through 23	2000
LSDSGU-50	1500
LSDSGU-75 through 100	1000
LSDSGU-200 through 400	800

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:58Z
Check the source to verify that this is the current version before use.

Custodians:
Army - MI
Navy - SH

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

Preparing Activity:
Navy - SH
(Project 6145-2308-013)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:58Z
Check the source to verify that this is the current version before use.